only be performed over IPv6.

These optimizations reduce both the size and the number of multicast packets, which is particularly beneficial on Wi-Fi networks. A Matter device that only supports IPv6 gets these optimizations automatically, simply by virtue of not supporting IPv4 at all.

For Thread mesh networks, where excessive use of multicast would be detrimental [RFC 7558], DNS-SD uses Unicast DNS instead, leveraging capabilities of the Thread Service Registry on the Thread Border Router [draft-lemon-stub-networks].

Conceptually, the DNS-SD [RFC 6763] information being communicated is identical to when Multicast DNS [RFC 6762] is being used, except that the information is communicated in unicast packets to and from a designated Service Registry, rather than being communicated in multicast packets to and from every other Node in the same broadcast domain.

Using Service Registration Protocol [SRP] and an Advertising Proxy [AdProx] running on the Thread Border Router, Matter Nodes on a Thread mesh can be discovered by other Matter Nodes on an adjacent Ethernet or Wi-Fi link, without the cost of using multicast on the Thread mesh. All Thread-connected Matter Nodes SHALL implement Service Registration Protocol.

Thread Border Routers advertise available SRP servers in the Thread Network Data [Thread specification]. Thread devices SHALL register their services using an available SRP server [Thread specification].

When Matter Nodes issue short-lived requests to other Matter Nodes, the response is sent back to the source IPv6 address and port of the request. When Matter Nodes issue long-lived requests to other Matter Nodes, by the time the response is generated the requester may have changed IPv6 address or port, so the responder may have to discover the current IPv6 address and port of the initiator in order to send the response.

A Thread Border Router SHALL implement DNS-SD Discovery Proxy [RFC 8766] to enable clients on the Thread mesh (e.g., other Nodes) to discover services (e.g., Matter Nodes) advertised using Multicast DNS on an adjacent Ethernet or Wi-Fi link, also without the cost of using multicast on the Thread mesh [draft-lemon-stub-networks]. For short-lived instantaneous queries, these queries can be performed using unicast DNS over UDP to the DNS-SD Discovery Proxy. For long-lived queries with ongoing change notification, DNS Push Notifications [RFC 8765] with DNS Stateful Operations [RFC 8490] allows clients on the Thread mesh to be notified of changes to the set of discovered services without expensive polling.

In principle, the Thread mesh Service Registry can be run on any capable Node(s) within (or even outside) the Thread mesh, though in practice the Thread Border Router is an attractive candidate to offer the Service Registry. Thread Border Router devices are typically AC-powered, and typically have more capable CPUs with greater flash storage and RAM than more constrained battery-powered Thread Nodes. Matter devices on Thread are dependent on Thread providing reliable service for those Thread devices on the Thread mesh. This is similar to how Matter devices on Wi-Fi are dependent on the Wi-Fi access point (AP) providing reliable service for those Wi-Fi devices using that Wi-Fi access point.